

Introduction to Algorithms

L01 - Algorithmic thinking

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For a given problem:

- Break the problem down into smaller constituent parts - **decomposition**
- Identify similar parts previously defined or from experience - **pattern recognition**
- Identify the steps required towards solving the problem - **abstraction**
- Define rules/steps that need to be applied - **algorithm**

Boiling the kettle

You decided to visit your grandmother the weekend before coming to UP to study COS. Your grandmother is an avid tea drinker and has a multitude of different types and flavours. She asks you to put the kettle on in order to make a cup of tea. As you are not going to be around that much, she asks you to “teach” your cousin who has moved in with her how to boil the kettle.

Boiling the kettle

How will you

- decompose the problem,
- apply patterns from previous experience,
- abstract the “act” of boiling the kettle into steps, and
- provide an algorithm to your cousin that will cover all possibilities

for boiling the kettle

Boiling the kettle

Escaping the canyon

Two hikers are dropped off individually anywhere in a narrow canyon that has a north-south orientation. The canyon goes on indefinitely in both directions. The only way for the hikers to escape from the canyon is to find each other and as a team climb out with their backpacks. The hikers may only perform the following actions when in the canyon: drop pack, pickup pack, walk north and walk south. Explain the best course of action the hikers should take in order for them to get out of the canyon.

Escaping the canyon

Apply the algorithmic thinking process to write an algorithm to escape the canyon.

Escaping the canyon

Before the next lecture, look for tools that could be used to model algorithmic thinking.